

HeartcareGPT: A Unified Multimodal ECG Suite for Dual Signal–Image Modeling and Understanding

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Abstract

*Although electrocardiograms (ECG) play a dominant role in cardiovascular diagnosis and treatment, their intrinsic data forms and representational patterns pose significant challenges for medical multimodal large language models (Med-MLLMs) in achieving cross-modal semantic alignment. To address this gap, we propose **Heartcare Suite**, a unified ECG suite designed for dual signal–image modeling and understanding: (i) **Heartcare-400K**. A fine-grained ECG instruction dataset on top of our data pipeline engine—**HeartAgent**—by integrating high quality clinical ECG reports from top hospitals with open-source data. (ii) **Heartcare-Bench**. A systematic benchmark assessing performance of models in multi-perspective ECG understanding and cross-modal generalization, providing guidance for optimizing ECG comprehension models. (iii) **HeartcareGPT**. Built upon a structure-aware discrete tokenizer **Beat**, we propose **Dual Stream Projection Alignment (DSPA)** paradigm—a dual encoder projection alignment mechanism enabling joint optimizing and modeling native ECG signal–image within a shared feature space. HeartcareGPT achieves consistent improvements across diverse ECG understanding tasks, validating both the effectiveness of the unified modeling paradigm and the necessity of a high-quality data pipeline, and establishing a methodological foundation for extending Med-MLLMs towards physiological signal domains. Our project is available at <https://github.com/ZJU4HealthCare/HeartcareGPT>.*

1. Introduction

Multimodal large language models (MLLMs) [2, 4, 6, 11, 21, 27, 33, 36, 43, 50, 51] demonstrate strong performance in general-purpose scenarios by jointly modeling multiple

modalities such as text, image and video. Motivated by their success in open-domain scenarios, researchers have extended the MLLM paradigm to the medical domain, giving rise to medical multimodal large language models (Med-MLLMs), including LLaVA-Med [20], HuatuoGPT-Vision [10], HealthGPT [25], and Lingshu [48]. These models show potential for medical diagnosis and reasoning, driving progress in intelligent healthcare applications.

In fact, existing Med-MLLMs are constrained by vision-centric paradigms that focus on imaging data [30, 56], yet medical applications frequently require integration of non-imaging modalities and domain-specific knowledge [42]. Electrocardiograms (ECG) illustrates this limitation, as it encodes critical diagnostic information in a dual signal–image form, tightly combining temporal and morphological features [5, 47]. Consequently, ECG not only delineates the failure boundary of existing approaches but also highlights the fundamental rationale for developing dedicated Med-MLLMs: *the structural characteristics of medical modalities must serve as the foundation of model design, rather than an afterthought for adaptation*.

Intelligent ECG interpretation relies on symbolic semantics to integrate patient history, ECG data, and clinical reports into interpretable reasoning chains. This paradigm, naturally suited to MLLMs, still faces fundamental challenges:

Data Misalignment. Mainstream ECG datasets such as PTB-XL [46] are designed for rhythm and disease classification, with annotations at the global level and lacking step-wise supervision. As a result, these datasets mainly enable direct mapping from ECG signals to labels, making it difficult to support interpretable reasoning or cross-modal information integration, and thus limiting the shift from simple classification to deeper understanding and reasoning.

Evaluation Deficiency. Existing medical benchmarks primarily emphasize classification accuracy and tend to overlook the assessment of comprehensive reasoning processes [29, 44]. For instance, they rarely evaluate models’ capabilities in clinical reasoning or pathology question–answering (QA) for ECG tasks. Therefore, it is essential

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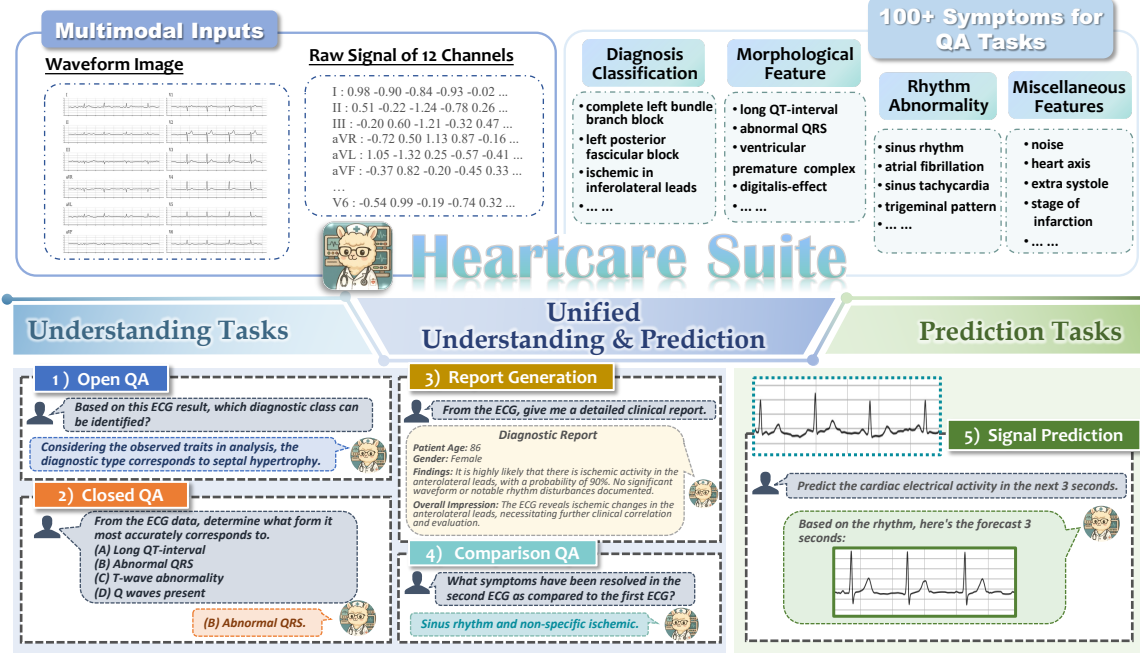


Figure 1. The proposed **Heartcare-400K** dataset. Heartcare-400K aggregates real-world ECG data, supporting *Closed-QA*, *Open-QA*, *Comparison-QA*, *Report Generation* and *Signal Prediction*.

to develop multidimensional assessment frameworks that include reliability and scenario-based metrics for evaluating the clinical utility of ECG-specific Med-MLLMs.

Dual Signal-Image Fusion Incapacity. The dual nature of ECG data—comprising both raw temporal signals and waveform images—poses unique modeling challenges [37, 41]. While signals capture dynamic cardiac activity and images reflect spatial morphology, both are essential for diagnosis. Most vision-focused MLLMs, however, process only ECG images [3, 31], neglecting key temporal features in the signal, such as intervals, calibration, inter-lead phase, and beat-to-beat changes. Moreover, multi-lead synchronous acquisition results in a tightly coupled, heterogeneous structure across temporal and spatial domains [23]. Addressing this dual complexity requires Med-MLLMs capable of consistent multimodal modeling and alignment.

We propose **Heartcare Suite**—a systematic and innovative framework for ECG, dedicated to establishing a unified and extensible ecosystem for ECG-specific Med-MLLMs:

(i) Dataset. We construct **Heartcare-400K**, a large-scale, fine-grained, multi-task multimodal ECG instruction dataset. It combines two sources: the public PTB-XL dataset [46] with 21,799 12-lead ECG signals annotated with 179 SCP-ECG classes, and 12,170 ECG images with structured reports from top hospitals, including scanned traces, clinical conclusions, and de-identified metadata—substantially enriching modality and label diversity. To transform heterogeneous ECG data into structured annotations, we develop **HeartAgent**, a multimodal engine with a bottom-up pipeline that

ensures annotation consistency and generates high-quality instruction-style QA pairs.

(ii) Benchmark. We propose **Heartcare-Bench**, the first fine-grained, multidimensional evaluation framework for ECG diagnostic intelligence, designed to assess a spectrum of model capabilities ranging from feature recognition to reasoning. Built upon Heartcare-400K, Heartcare-Bench systematically covers five major task types—*Closed-QA*, *Open-QA*, *Comparison-QA*, *Report Generation*, and *Signal Prediction*—spanning key diagnostic dimensions such as rhythm, waveform, and morphology. It comprises three complementary modality subsets: *Signal (S)*, *Image (I)*, and *Cross-Modal (C)*, enabling unified evaluation from single-modality reasoning to multi-ECG semantic alignment. With a hierarchical, multi-metric scoring system, Heartcare-Bench integrates knowledge reasoning and cross-modal understanding within a unified evaluation coordinate.

(iii) Model. The dual-form characteristics of ECG introduce unique structural complexity in modeling. We propose **HeartcareGPT**, aiming to build ECG-specific Med-MLLMs. We design **Bidirectional ECG Abstract Tokenization (Beat)**, a structure-aware discrete encoding mechanism centered on vector quantization [12, 45, 54], which maps high-frequency continuous signals into token sequences. The design comprises three components: (i) *Dual-level Vector Quantization (DVQ)*, which refines rhythm and inter-lead phase dependencies captured by the codebook to achieve high-fidelity compression; (ii) *Query-guided Bidirectional Diffusion (QBD)*, which jointly models past and future con-

texts within the latent token space to support both signal reconstruction and prediction; and (iii) *Joint Supervision Strategy*, which jointly optimizes reconstruction and prediction to maximize clinical semantic fidelity during encoding. Furthermore, we propose **Dual Stream Projection Alignment (DSPA)**, which employs dual experts to separately process ECG inputs. Through distinct preprocessing strategies and modality-specific encoders, ECG representations are transformed into embeddings compatible with Med-MLLMs. All modality embeddings are projected into a shared language space and concatenated into a unified sequence, enabling cross-modal joint reasoning for ECG under a unified autoregressive paradigm.

Experimental results demonstrate the powerful paradigm of Heartcare Suite. The main contributions of this work are as follows:

- **High-quality ECG Instruction Dataset.** Heartcare-400K serves as the first comprehensive ECG instruction dataset, which significantly enhances Med-MLLM performance across ECG-related tasks.
- **Multidimensional ECG Benchmark.** We propose Heartcare-Bench, an evaluation framework that assesses clinical performance of ECG tasks for Med-MLLMs.
- **Fine-grained ECG Understanding Paradigm.** We develop HeartcareGPT, the first model supporting pathology-level ECG understanding with state-of-the-art (SOTA) results.

2. Related Work

Multimodal Representation Learning for ECG. Recent advances in multimodal ECG representation learning follow three main directions. First, signal-semantic alignment. ECG-SL [52] and MERL [26] align heartbeats and clinical reports via self-supervision and knowledge prompting, while HeartLang [19] decomposes waveforms into semantic tokens for fine grained cardiac analysis. Second, cross-lead fusion. ECG-DAN [9] adopts a dual attention network to balance global cross lead interactions with local temporal dynamics, and ESI [53] adds a signal text contrastive learning to strengthen robustness under limited labels. Third, large language model (LLM)-driven pretraining. ECG-LM [49] maps ECG embeddings into a pretrained language space, and SuPreME [7] structures domain knowledge from clinical reports for pretraining. However, existing approaches typically address isolated aspects such as reconstruction quality, lead interaction, or semantic alignment, without forming a unified end-to-end multimodal ECG modeling framework.

Medical Multimodal Large Language Models. Med-MLLMs have shown strong capabilities in medical understanding and diagnostic support [10, 18, 35, 48]. Med-Flamingo [29] and LLaVA-Med [20] are representative early medical multimodal models, focusing on image–text alignment and visual question answering. MedVLM [35]

employs multi-stage pretraining and attains SOTA performance in radiology report generation and organ localization. HealthGPT [25] unifies image understanding and generation within one framework. Domain-specific variants—LLaVA-Rad [8], EyecareGPT [22], and SkinGPT-4 [57]—enable structured reporting and multimodal reasoning across radiology, ophthalmology, and dermatology. However, the dual-form characteristics of ECG presents significant challenges to vision-centric models, and an effective ECG-specific Med-MLLM framework remains absent to date.

3. Heartcare Suite: Heartcare-400K

3.1. Data Collection and Organization

Existing ECG datasets suffer from limited modalities, coarse annotations, insufficient scale, and substantial heterogeneity in signal sampling, lead configuration, and preprocessing, all of which hinder the development of Med-MLLMs for intelligent ECG diagnosis. To address these gaps, we introduce **Heartcare-400K**, a large-scale multimodal ECG QA dataset with two complementary modalities: (i) **structured digital signals** and (ii) **unstructured ECG report images**. We normalize them by rendering all signals into a unified ECG visual format consistent with clinical layout. These visualized signals, together with native report images, enrich modality diversity and support comprehensive ECG understanding.

To construct Heartcare-400K, we collaborated with several major public hospitals to collect 12,170 standardized PDF-format clinical ECG reports. These reports include patient demographics, physiological parameters, physician diagnoses, and about 5 seconds of 12-lead waveform images. In addition, we systematically integrated multiple publicly available digital signal ECG datasets, with PTB-XL [46] as the primary source. PTB-XL is one of the largest open ECG repositories, containing 21,799 12-lead records sampled at 500 Hz over 10 seconds, with standardized diagnostic labels and detailed patient metadata (such as gender, age and weight).

Raw ECG data typically contain only brief diagnostic text and lack the instruction-style supervision and semantic diversity needed for MLLM fine-tuning. To address this, we developed a multimodal data engine for automated extraction, cleaning, standardization, and expert review, and further augmented the dataset with QA pairs from the publicly available ECG-QA [32], which supports both single-modality queries and cross-modality reasoning tasks.

Ultimately, Heartcare-400K is organized into four types of QA tasks: (i) *Closed-QA* (single-ECG multiple-choice question), (ii) *Open-QA* (single-ECG short-form question), (iii) *Comparison-QA* (multi-ECG multiple-choice question), (iv) *Report Generation* (long-form answers), and (v) *Signal Prediction* (ECG generation). As Figure 1 shows, these tasks

equip models with fine-grained ECG comprehension and clinical reasoning capabilities, making Heartcare-400K a foundational resource for developing practical and generalizable intelligent ECG diagnosis systems.

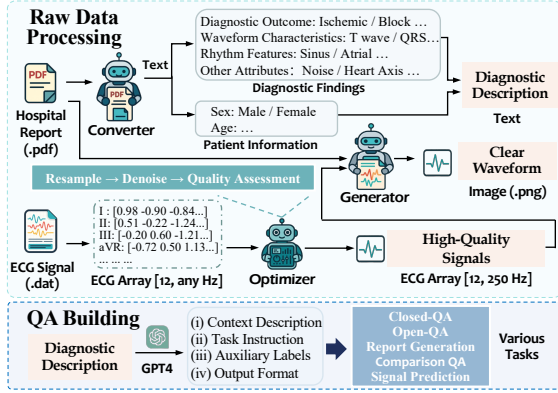


Figure 2. Framework of HeartAgent for QA generation.

3.2. Multimodal Data Engine

To efficiently construct Heartcare-400K, we design **HeartAgent** (as shown in Figure 2), an automated multimodal data engine with two key stages: *raw data processing* and *data building*. Details of each module are provided in the Appendix A.

Raw Data Processing. This stage integrates three core components to transform heterogeneous ECG sources into standardized inputs: (i) **Multimodal Feature Converter**. Extracts patient information and clinical descriptions from hospital PDFs, maps diagnostic text to structured labels, and prepares both text and image data for subsequent processing. (ii) **Noise Filtering and Quality Optimizer**. Unifies signal formats and applies denoising, resampling, and quality assessment to produce high-quality, standardized ECG signals. (iii) **Diversified Image Generator**. Renders clean waveform images from processed digital signals and generates de-identified tracings from hospital reports, ensuring multi-modal data alignment.

Data Building. This stage is handled by the **Multi-task QA Builder**, which constructs instruction-style QA samples using GPT-4 [33]. Each sample contains structured context, clear task instructions, auxiliary labels, and standardized answers, supporting four diverse ECG understanding tasks and a *Signal Prediction* task critical for comprehensive model training.

These components collectively ensure Heartcare-400K achieves high-quality, diverse, and task-relevant multimodal annotations.

4. Heartcare Suite: Heartcare-Bench

To systematically evaluate the capabilities of Med-MLLMs across unified ECG understanding and prediction tasks, we

propose Heartcare-Bench—a multidimensional benchmark encompassing five major task types: *Closed-QA*, *Open-QA*, *Comparison-QA*, *Report Generation*, and *Signal Prediction*. Heartcare-Bench is structured into three complementary subsets based on input modality: (i) **Heartcare-Bench^S** (signal-based), (ii) **Heartcare-Bench^I** (image-based), and (iii) **Heartcare-Bench^C** (cross-modal comparison). In addition, Heartcare-Bench employs strict patient-level data partitioning with thorough cross-split duplicate inspection; full details are provided in Appendix B.4.

Heartcare-Bench^{SI}: Single-Modality Subsets. The signal and image subsets share a unified framework to evaluate four core clinical dimensions: *Diagnosis*, *Waveform analysis*, *Rhythm interpretation* and *Miscellaneous Features*. To broaden task diversity and clinical relevance, we further integrate QA pairs from ECG-QA [32], primarily those derived from PTB-XL, enriching *Closed-QA* and *Open-QA* settings and enabling categorization by these four dimensions. The *Miscellaneous (Misc.)* category further assesses ECG attributes such as noise, infarction stage, ectopic beats, and cardiac axis. In addition to QA tasks, the single-modality subsets also include *Report Generation* and *Signal Prediction* tasks. Detailed formats and examples are provided in Appendix B.4.

Heartcare-Bench^C: Cross-Modality Comparison Subset. Beyond single-modality tasks, this subset evaluates a model’s ability to reason over multiple ECGs through comparison QA tasks adapted from ECG-QA [32]. It is centered around three modality comparison dimensions: (i) *signal-signal (S-S)*, (ii) *image-image (I-I)*, and (iii) *signal-image (S-I)*. Each configuration is further categorized into two clinically relevant subtypes: consecutive (ECGs from the same patient, assessing temporal consistency) and irrelevant (ECGs from different patients, evaluating differential diagnostic reasoning).

Overall, Heartcare-Bench adopts a multidimensional evaluation protocol covering semantic consistency, linguistic fluency, clinical accuracy, and waveform prediction quality. Full scoring rules and evaluation details are provided in Appendix B.4.

5. Methodology

5.1. Bidirectional ECG Abstract Tokenization

We present a structure-aware discrete encoding mechanism centered on vector quantization—**Bidirectional ECG Abstract Tokenization (Beat)**. The architecture of Beat is depicted in Figure 3 (a).

Forward Diffusion Process. Given an acquired ECG signal, we denoise and resample it to obtain a representative segment $\mathbf{S} \in \mathbb{R}^{T \times C}$ (sequence length T , C leads). We then partition $\mathbf{S}$ into contiguous, non-overlapping temporal patches and map them with a projection layer Ψ to yield the

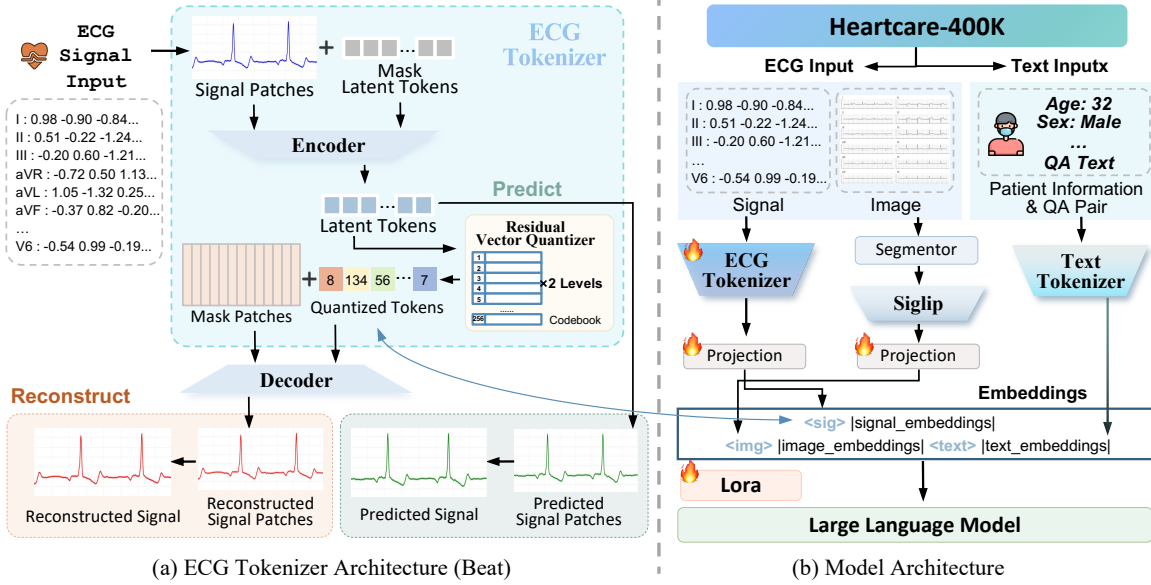


Figure 3. Architecture of HeartcareGPT. (a) The dual-form ECG inputs are routed and encoded with modality-specific expert projections aligned to the LLM backbone. (b) The unified autoregressive architecture efficiently supports interleaved and joint modeling of ECG multimodal inputs.

corresponding patch embeddings:

$$\mathbf{e} = \Psi(\text{Patchify}_f(\mathbf{S}) \mid \theta_\Psi) \in \mathbb{R}^{t \times c}, t = \frac{T}{f}, c = Cf, \quad (1)$$

where f is the patch frame size, t the number of patches, and θ_Ψ the projection parameters. To inject high-level semantic control, we append m learnable queries $\mathbf{q} \in \mathbb{R}^{m \times c}$ to the patch embeddings $\mathbf{e}$, forming the model input $\mathcal{H}_{\text{in}} = [\mathbf{e} \parallel \mathbf{q}]$. A Transformer encoder then performs forward diffusion, yielding compressed contextual representations:

$$\mathcal{H}_{\text{latent}}^q = (\text{TransformerEnc}(\mathcal{H}_{\text{in}}))_{[t:t+m]} \in \mathbb{R}^{m \times c}. \quad (2)$$

Dual-level Vector Quantization. To compress ECG signals while preserving rhythmic structure and salient pathological cues, we adopt dual-level VQ with a core codebook $\mathcal{C}_1$ and a residual codebook $\mathcal{C}_2$. For each query vector $\mathbf{h}_q^i = (\mathcal{H}_{\text{latent}}^q)_{[i,:]} \in \mathbb{R}^c$, we first apply the core quantization:

$$\hat{\mathbf{h}}_{(q,1)}^i = \text{Quant}_{\mathcal{C}_1}(\mathbf{h}_q^i) = \arg \min_{\mathbf{c} \in \mathcal{C}_1} \|\mathbf{h}_q^i - \mathbf{c}\|_2. \quad (3)$$

The residual is then quantized by the codebook $\mathcal{C}_2$:

$$\hat{\mathbf{h}}_{(q,2)}^i = \text{Quant}_{\mathcal{C}_2}(\mathbf{h}_q^i - \hat{\mathbf{h}}_{(q,1)}^i) = \arg \min_{\mathbf{c} \in \mathcal{C}_2} \|\mathbf{h}_q^i - \hat{\mathbf{h}}_{(q,1)}^i - \mathbf{c}\|_2. \quad (4)$$

Finally, we quantize the forward-diffused representation to $\hat{\mathbf{h}}_{\text{latent}}^i = \hat{\mathbf{h}}_{(q,1)}^i + \hat{\mathbf{h}}_{(q,2)}^i$, yielding a high-fidelity mapping from continuous latents to a discrete code space and significantly improving reconstruction quality.

Query-guided Bidirectional Diffusion. Forward diffusion (Eq. 2) compresses the ECG into a dense discrete space

but offers no guarantees of completeness or invertibility. We therefore adopt an autoencoding view and introduce a reverse diffusion anchored at the quantized query $\mathcal{H}_{\text{latent}}^q$, casting compression and reconstruction as a joint constraint: the forward path enforces compact coding, the reverse path restores details, yielding a closed-loop, bidirectional token refinement between latent and signal spaces.

During reverse diffusion, the original input $\mathbf{e}$ is mapped to its forward-diffusion slot $\mathcal{H}_{\text{origin}}^q$ and masked by $\mathcal{M}$ to prevent leakage, so only the query vectors $\mathcal{H}_{\text{latent}}^q$ retain ECG signal information for reconstruction:

$$\mathcal{H}_{\text{out}}^{\text{recon}} = (\text{TransformerDec}(\mathcal{H}_{\text{latent}}^q; \mathcal{M}))_{[0:t]}. \quad (5)$$

Joint Supervision Strategy. To fully exploit the bidirectional diffusion capacity, Beat employs a multi-objective loss that jointly optimizes reconstruction and compression, integrating three objectives. The reconstruction and prediction losses are defined as:

$$\mathcal{L}_{\text{recon}} = \|\mathcal{H}_{\text{out}}^{\text{recon}} - \mathbf{e}\|_2, \quad \mathcal{L}_{\text{pred}} = \|\mathcal{H}_{\text{latent}}^q - \mathbf{e}_{\text{pred}}\|_2. \quad (6)$$

where $\mathbf{e}_{\text{pred}}$ denotes the ECG segment used for prediction. The vector quantization loss is defined as:

$$\mathcal{L}_{\text{VQ}} = \sum_{i,j} \|\text{sg}[\mathbf{h}_i^i] - \hat{\mathbf{h}}_{(q,i)}^i\|_2^2 + \beta \|\mathbf{h}_j^i - \text{sg}[\hat{\mathbf{h}}_{(q,j)}^i]\|_2^2, \quad (7)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operation, and $\mathbf{h}_j^i$ refers to the feature vector before quantization at the j -th level. The overall training objective is given by $\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{recon}} + \lambda_2 \mathcal{L}_{\text{pred}} + \lambda_3 \mathcal{L}_{\text{VQ}}$.

Table 1. Performance comparison between HeartcareGPT and other baselines on *Closed-QA* tasks from Heartcare-Bench^S and Heartcare-Bench^I, evaluated by accuracy. We use **bold** text to indicate the best results and underline to indicate the second-best results.

Model	Heartcare-Bench ^S				Heartcare-Bench ^I				Avg.
	Diagnosis	Waveform	Rhythm	Misc.	Diagnosis	Waveform	Rhythm	Misc.	
Generalist Models									
LLaVA-1.5-7B [27]	19.26	27.03	22.19	28.96	15.15	12.40	18.20	25.47	21.08
Qwen2.5-VL-7B [6]	28.29	35.36	31.70	35.31	35.84	31.06	22.35	41.41	32.67
InternVL-2.5-8B [11]	29.87	34.18	31.25	40.75	25.86	17.89	38.23	40.05	32.26
Yi-VL-6B [51]	49.19	38.90	31.06	39.65	59.55	44.46	17.06	39.85	39.97
MiniCPM-V2.6-8B [14]	32.37	31.34	23.40	32.23	23.71	33.59	10.05	30.63	27.17
Gemma-3-4B [43]	23.84	24.36	18.30	21.47	38.73	38.09	20.63	19.32	25.59
Claude-3.5 [2]	37.37	20.73	15.96	35.16	30.58	37.76	37.83	34.45	31.23
GPT5 [34]	29.87	15.23	19.57	25.23	41.42	34.14	33.73	26.23	28.30
Medical Models									
Lingshu-7B [48]	29.56	33.44	40.00	20.75	47.96	37.98	19.31	30.28	32.41
LLaVA-Med-7B [20]	27.79	25.86	24.04	33.69	25.11	25.96	26.17	33.36	27.75
MedVLM-R1-2B [35]	16.27	15.42	15.32	36.32	14.08	19.43	8.33	35.56	20.09
HealthGPT-M3-3.8B [25]	21.63	10.87	26.58	20.23	15.35	12.41	16.06	23.97	18.39
HeartcareGPT-3.8B	81.95	95.94	82.79	79.84	87.85	92.21	79.25	67.80	83.33
HeartcareGPT-7B	86.16	98.62	93.20	75.15	62.71	87.82	85.13	78.59	83.42

Tokenization. During inference, we discard the decoder and prediction heads, retaining only the encoder and quantizer. Given an ECG segment $\mathbf{S}$ of length T , Beat applies temporal normalization: if $T > t$, split into $\lceil T/t \rceil$ slices from right to left; if $T < t$, left-pad to t . For recordings with fewer than 12 leads, pad missing channels with zeros. Each segment is then encoded into discrete tokens:

$$\mathbf{S} = \text{Beat}(\mathbf{S}) = \{c_1^1, \dots, c_1^n \| c_2^1, \dots, c_2^n\}, c_i^j \in \{\mathcal{C}_1, \mathcal{C}_2\}. \quad (8)$$

These discrete tokens can be directly used as multimodal extensions of LLM vocabulary, enabling unified semantic modeling and cross-modal reasoning between ECG signals and texts.

For autoregressive forecasting, Beat predicts the next t' segment conditioned on the previous context $(T + t')$, iteratively generating arbitrary-length ECG sequences.

5.2. HeartcareGPT

As Figure 3 (b) shows, we present **HeartcareGPT**, which maps ECG signals, ECG images, and text into a unified discrete space, enabling ECG reasoning with a single autoregressive architecture.

Unified ECG Tokens. Specifically, for a multi-lead ECG $\mathbf{S} \in \mathbb{R}^{T \times C}$, Beat (Eq. 8) encodes it into a token sequence $\mathbf{S}$ capturing elementary ECG morphology. For ECG images, we partition and rearrange them into 12 lead-wise maps to decouple lead semantics, then apply SigLip [55] to obtain visual tokens $\mathcal{V}$ that embed per-lead waveform and spatial layout. Meanwhile, clinical notes, basic physiological data, and diagnostic instructions are tokenized by the native tokenizer of the LLM into textual features $\mathcal{F}_T$, providing a semantic anchor for non-text modalities and enabling fine-grained alignment and joint modeling in the shared token space.

Dual Stream Projection Alignment. To avoid parameter interference among modalities in shallow layers, we adopt a decoupled design, mapping sequential feature $\mathcal{F}_S$ and visual feature $\mathcal{F}_V$ into the model’s embedding space through dedicated expert projections Ψ_{sig} and Ψ_{img} :

$$\mathcal{F}_S = \Psi_{\text{sig}}(\mathcal{S} | \theta_{\Psi_{\text{sig}}}), \quad \mathcal{F}_V = \Psi_{\text{img}}(\mathcal{V} | \theta_{\Psi_{\text{img}}}), \quad (9)$$

so that they lie in the same representation space. Either projection layer is implemented as a lightweight MLP block. Subsequently, at the input side we directly concatenate $\mathcal{F}_S$, $\mathcal{F}_V$ and $\mathcal{F}_T$ into a single long sequence:

$$\langle \text{BOS} \rangle \langle \text{sig} \rangle \{ \mathcal{F}_S \} \langle \text{img} \rangle \{ \mathcal{F}_V \} \langle \text{text} \rangle \{ \mathcal{F}_T \} \langle \text{EOS} \rangle, \quad (10)$$

where $\langle \text{sig} \rangle$ and $\langle \text{img} \rangle$ are specially introduced conditional tokens that can be expanded as needed at the implementation level to accommodate multiple signal segments and multiple images. With this design, the routing and composition of multimodal inputs no longer rely on additional architectural components, but are uniformly reduced to simple template filling of instruction prompts.

Training Strategy. To enable stable coexistence of tokens from heterogeneous modalities and distributions within the base model, HeartcareGPT adopts an overall training pipeline with clearly separated roles: **(i) Multimodal warm-up**, where we only train the signal and image projection layers using a captioning task to preliminarily align multimodal features with textual features; **(ii) Joint fine-tuning**, where, once all modalities have been embedded into a compatible space, we unfreeze the weights of the LLM and optimize it together with the modality projection layers in an end-to-end manner on instruction tasks.

Autoregressive Multimodal Generation. The unified training objective of the model is to generate the corresponding textual output $\mathcal{R} = [r_1, r_2, \dots, r_{N_r}]$ conditioned on the

Table 2. Performance comparison between HeartcareGPT and baselines on *Open-QA* tasks from Heartcare-Bench^S and Heartcare-Bench^I.

Model	Heartcare-Bench ^S								Heartcare-Bench ^I							
	Diagnosis		Waveform		Rhythm		Misc.		Diagnosis		Waveform		Rhythm		Misc.	
	F1-Bio	Rouge-L	F1-Bio	Rouge-L	F1-Bio	Rouge-L	F1-Bio	Rouge-L	F1-Bio	Rouge-L	F1-Bio	Rouge-L	F1-Bio	Rouge-L	F1-Bio	Rouge-L
<i>Generalist Models</i>																
LLaVA-1.5-7B [27]	18.52	11.57	25.34	13.93	20.84	12.45	27.53	14.82	14.82	10.25	11.24	9.67	17.53	11.38	24.21	13.76
Qwen2.5-VL-7B [6]	27.63	12.84	34.15	15.72	30.82	13.96	34.27	16.38	35.24	14.93	30.35	13.27	21.84	12.18	40.13	17.24
InternVL-2.5-8B [11]	29.27	13.29	33.52	14.86	30.64	13.72	39.82	16.95	25.37	11.82	17.26	10.45	37.53	15.63	39.27	16.84
Yi-VL-6B [51]	48.53	15.38	38.27	14.72	30.42	13.85	38.96	15.63	58.92	17.82	43.85	16.27	16.43	10.84	38.74	15.49
MiniCPM-V2.6-8B [14]	31.84	12.93	30.73	12.84	22.86	11.75	31.52	13.92	23.17	11.28	32.94	13.46	9.53	8.67	29.83	13.27
Gemma-3-4B [43]	12.90	5.49	11.57	6.95	13.85	5.87	23.01	6.08	14.13	6.65	14.74	9.90	15.12	5.95	21.02	8.98
Claude-3.5 [2]	21.43	5.04	14.81	4.36	23.01	6.75	17.23	7.91	21.43	5.39	27.69	5.02	17.78	4.13	19.14	10.24
GPT5 [34]	23.00	7.25	21.42	13.93	37.39	14.71	24.83	9.57	40.82	15.93	33.52	12.84	33.17	12.53	25.64	9.82
<i>Medical Models</i>																
Lingshu-7B [48]	28.94	12.37	32.84	14.26	39.27	15.84	20.17	10.53	47.36	16.28	37.35	15.12	18.74	10.87	29.63	13.15
LLaVA-Med-7B [20]	27.15	12.84	25.27	11.93	23.48	11.27	33.02	14.38	24.53	11.29	25.37	12.15	25.53	11.84	32.74	14.26
MedVLM-R1-2B [35]	15.64	8.73	14.83	8.25	14.73	8.16	35.64	13.52	13.52	7.84	18.84	9.27	16.83	6.45	34.92	13.17
HealthGPT-M3-3.8B [25]	21.07	10.82	10.27	8.93	25.94	11.76	19.63	10.24	14.83	9.67	11.84	8.75	15.43	9.82	23.37	11.39
HeartcareGPT-3.8B	68.53	32.27	72.74	35.38	78.63	36.42	65.84	30.97	63.17	29.83	70.92	33.57	75.36	34.69	61.53	28.24
HeartcareGPT-7B	72.94	36.85	78.26	39.64	73.62	35.27	68.37	34.18	67.43	33.52	76.73	37.84	69.84	32.47	64.17	31.73

above multimodal input $\mathcal{U} = \{\mathcal{F}_S, \mathcal{F}_V, \mathcal{F}_T\}$, by maximizing the following probability distribution:

$$P_\theta(\mathcal{R} | \mathcal{U}) = \prod_{j=1}^{N_r} P_\theta(r_j | \{\mathcal{U} | r_{<j}\}). \quad (11)$$

Here, θ denotes the parameters of $\mathcal{M}_{\text{llm}}$. The above optimization equips the model with strong capabilities in ECG diagnosis and cross-modal knowledge complementation, yielding a general paradigm for ECG specific Med-MLLMs.

6. Experiments

6.1. Data and Experimental Setup

Data Details. We systematically evaluate our model with baseline models on the proposed Heartcare-Bench^S, Heartcare-Bench^I and Heartcare-Bench^C, providing a comprehensive assessment of generalization and diagnostic accuracy. For baseline models that cannot accept digital signal input directly, we convert signals into images. More details are provided in Appendix B.3.

Model Details. HeartcareGPT employs SigLip [55] and the proposed Beat as dual-form feature encoders and adopts a three-stage training paradigm: (i) training Beat to extract high-fidelity ECG embeddings, (ii) warming up the visual and signal projectors to stabilize feature alignment, and (iii) performing joint instruction fine-tuning on Heartcare-400K for end-to-end modeling. Detailed hyperparameter settings and model configurations are provided in Appendix B.2.

Baseline. We conduct a zero-shot evaluation on 12 representative LLMs, including eight open-world LLMs (e.g., LLaVA-v1.5 [20], Qwen2.5-VL [6], InternVL2.5 [11], Yi-VL [51], MiniCPM-V2.6 [14], gemma-3 [43], Claude-3.5 [2], GPT5 [34] and four Med-MLLMs (e.g., Lingshu-7B [48], LLaVA-Med [20], MedVLM-R1 [35], HealthGPT [25]). *Signal prediction* tasks are not included in the evaluation when baseline models fail to respond to prediction instructions correctly. More details refer

to Appendix B.1.

6.2. Main Results

Closed-QA. As shown in Table 1, HeartcareGPT-7B achieves SOTA performance on *Closed-QA* with an average accuracy of 83.42%, while HeartcareGPT-3.8B achieves 83.33%, surpassing other models across all subtasks by a large margin. We attribute this to the ECG-aware tokenization and instruction tuning framework, which enables precise alignment between temporal signal patterns and clinically grounded language reasoning.

Open-QA. Table 2 reports the results on *Open-QA* tasks, evaluated using *BioBERTScore-F1 (F1-Bio)* [38] and *ROUGE-L* [24]. Our HeartcareGPT series achieve the highest overall performance across four subtasks, demonstrating strong capability in generating clinically relevant, semantically consistent answers grounded in ECG signals.

Comparison-QA. The *Comparison-QA* task assesses cross-modal reasoning by requiring analysis and contrast of two ECG inputs—either signals, images, or signal-image pairs—in Heartcare-Bench^C. As shown in Table 3, HeartcareGPT excels in this challenging setting, with both 3.8B and 7B models achieving SOTA performance. This success underscores the effectiveness of our multimodal fusion design, which enables a unified and nuanced understanding across heterogeneous ECG representations.

Report Generation. Table 4 presents HeartcareGPT’s *Report Generation* results on Heartcare-Bench^S and Heartcare-Bench^I, evaluated by *Score^{GPT}*, *F1-Rad* [17], and *ROUGE-L* [24]. HeartcareGPT achieves top scores on most metrics, outperforming all baseline models. While Qwen2.5-VL-7B scores higher on some metrics related to report structure and expression, HeartcareGPT excels in clinical content accuracy and reliability. Since GPT-based evaluation considers formatting and completeness, score differences often reflect output style rather than clinical accuracy. Full criteria and

Table 3. Performance comparison between HeartcareGPT and other baselines on *Comparison-QA* tasks from Heartcare-Bench^C. *Cons.* = Consecutive ECGs from the same patient; *Irr.* = Irrelevant ECGs from different patients. Yi-VL-6B and HealthGPT-M3-3.8B do not support multi-image input.

Model	S-S		I-I		S-I		Avg.
	Cons.	Irr.	Cons.	Irr.	Cons.	Irr.	
Generalist Models							
LLaVA-1.5-7B [27]	62.70	56.24	66.17	63.51	47.94	45.65	57.04
Qwen2.5-VL-7B [6]	50.64	47.26	49.36	50.68	50.21	51.28	49.91
InternVL-2.5-8B [11]	47.83	45.13	46.18	43.33	49.40	43.24	45.85
MiniCPM-V2.6-8B [14]	53.22	48.13	51.87	45.50	49.85	45.05	48.97
Gemma-3-4B [43]	49.18	38.53	49.63	35.44	49.55	35.59	42.99
Claude-3.5 [2]	46.17	43.96	39.15	56.78	24.47	54.82	44.23
GPT5 [34]	50.97	49.18	50.82	49.40	50.00	47.90	49.71
Medical Models							
Lingshu-7B [48]	58.79	45.01	62.13	59.46	45.11	43.12	52.27
LLaVA-Med-7B [20]	35.46	34.82	32.84	32.65	43.26	34.14	35.53
MedVLM-R1-2B [35]	58.44	39.33	53.69	40.34	49.57	37.95	46.55
HeartcareGPT-3.8B	66.40	67.47	69.88	75.19	78.71	78.83	72.74
HeartcareGPT-7B	74.01	75.87	74.80	79.09	80.05	79.55	77.23

model comparisons are detailed in Table 7.

Table 4. Performance comparison between HeartcareGPT and other baseline methods on *Report Generation* tasks from Heartcare-Bench^S and Heartcare-Bench^I.

Model	Heartcare-Bench ^S			Heartcare-Bench ^I		
	Score ^{GPT}	F1-Rad	Rouge-L	Score ^{GPT}	F1-Rad	Rouge-L
<i>Generalist Models</i>						
LLaVA-1.5-7B [27]	55.69	5.22	18.30	37.40	12.90	20.75
Qwen2.5-VL-7B [6]	64.60	5.64	9.00	67.40	7.67	13.56
InternVL-2.5-8B [11]	42.07	5.50	9.13	33.69	7.73	11.02
Yi-VL-6B [51]	26.58	3.75	13.05	21.74	6.73	15.18
MiniCPM-V2.6-8B [14]	34.54	5.74	10.66	49.60	7.48	12.63
Gemma-3-4B [43]	64.57	4.38	7.57	57.10	7.00	9.07
Claude-3.5 [2]	63.11	5.03	12.33	63.29	7.63	14.02
GPT5 [34]	62.73	4.85	8.86	78.80	6.29	9.55
<i>Medical Models</i>						
Lingshu-7B [48]	58.89	7.13	13.94	51.94	9.52	16.31
LLaVA-Med-7B [20]	50.02	5.21	14.95	27.71	6.56	16.45
MedVLM-R1-2B [35]	32.26	2.10	15.51	56.58	9.05	18.27
HealthGPT-M3-3.8B [25]	25.39	1.00	7.53	37.62	1.22	9.13
HeartcareGPT-3.8B	61.29	26.84	34.39	78.50	23.10	38.68
HeartcareGPT-7B	76.55	21.70	32.55	65.03	27.17	44.86

6.3. Ablation Study and Expert Evaluation

We conduct extensive ablation studies to analyze the contribution of each design component in HeartcareGPT. The complete results are presented in Appendix C.1.

The Three-Stage Training Pipeline. We further verify the necessity of each training stage in our three-phase optimization scheme. We experiment with models that omit (i) Step 1: beat-level training, (ii) Step 2: projector warming-up, and (iii) both steps simultaneously. Figure 4 show our results of ablation studies on the training pipeline. The performance significantly drops in all ablated variants, indicating that each stage plays an indispensable role in stabilizing multimodal alignment and improving diagnostic accuracy.

Multimodal Integration. To assess multimodal fusion, we compare our tri-modal (signal-image-text) model against single-modality baselines (image-only and signal-only). We

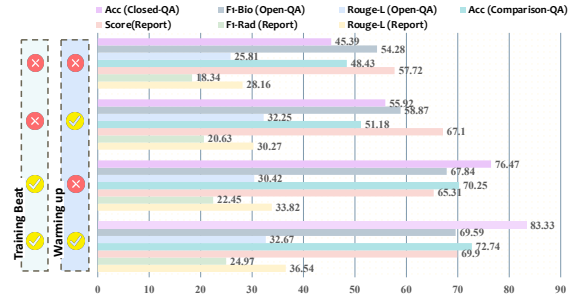


Figure 4. Results of ablation studies on training pipeline.

further examine ECG image segmentation by contrasting full-image and 12-lead sub-image strategies. As summarized in Figure 5 (a), multimodal fusion and sub-image segmentation yield clear performance gains, highlighting the value of integrating diverse ECG modalities for improved diagnostic understanding.

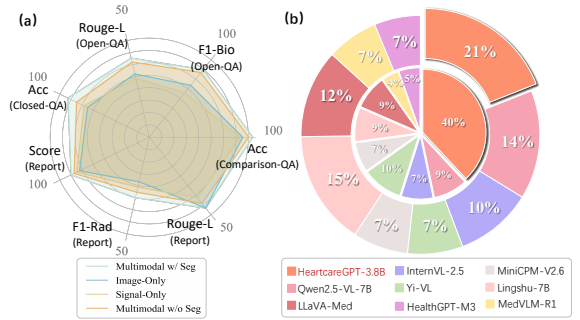


Figure 5. (a) Results of ablation studies on multimodal integration. (b) Expert preference distribution across models. Inner-ring results are based on *Open-QA* evaluations; outer-ring results are based on *Report Generation* evaluations.

Expert Evaluation. We conduct expert evaluation to assess clinical preference on *Open-QA* and *Report Generation* tasks. Ten board-certified cardiologists reviewed 400 sampled cases with shuffled outputs from HeartcareGPT-3.8B and eight representative baselines, selecting responses best aligned with clinical reasoning and diagnostic conventions. As shown in Figure 5 (b), HeartcareGPT achieves the highest first-choice selections—40% in *Open-QA* and 21% in *Report Generation*—substantially surpassing all baselines (7–15%). These results demonstrate clear clinical preference for HeartcareGPT, with expert scoring consistent with automated metrics, further supporting its clinical reliability.

7. Conclusion

Heartcare Suite establishes a comprehensive multimodal foundation framework for fine-grained ECG understanding, integrating high-quality dataset, clinically aligned benchmarks, and scalable modeling strategies. We hope this work serves as a stepping stone for future research on Med-MLLMs in clinically grounded signal-language reasoning.

Acknowledgements

This work has been supported in part by the NSFC (No. 62436007), the ZJNSF (No. LZ25F020004), the Key Research and Development Projects in Zhejiang Province (No. 2025C01128, 2025C01030, 2025C02156), Ningbo Yongjiang Talent Introduction Programme (2023A400-G).

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